

Module 4

Probabilistic Based Models

Probabilistic Based Models

- Probabilistic Generative Model-Naive Bayes: -Bayes' Theorem and Conditional Probability-Maximum Likelihood- Gaussian, Multinomial, and Bernoulli Naive Bayes. Inference in Bayesian Belief Network, Markov Models.

What is probabilistic model?

- A **probabilistic model** in Machine Learning is a model that uses **probability theory** to represent uncertainty in data and make predictions. Instead of giving only fixed outputs, it estimates the **likelihood of different possible outcomes**.
- Let's consider a classification problem with N classes. If the classification model (classifier) is probabilistic, for a given input, it will provide probabilities for each class (of the N classes) as the output.
- In other words, a probabilistic classifier will provide a probability distribution over the N classes.

It learns the probability distribution of data, such as:

$$P(Y|X)$$

where:

- X = input features
- Y = target/output

This means the model predicts the probability of output Y given input X .

Example:

For email classification:

- 90% probability = Spam
- 10% probability = Not Spam

Instead of only saying "Spam," it gives confidence levels.

Probabilistic Model Vs Non probabilistic

-An example....

- Take the task of classifying an image of an animal into five classes — {Dog, Cat, Deer, Lion, Rabbit} as the problem.
- As input, we have an image (of a dog).
- For this example, let's consider that the classifier works well and provides correct/ acceptable results for the particular input we are discussing.
- When the image is provided as the input to the probabilistic classifier, it will provide an output such as (Dog (0.6), Cat (0.2), Deer(0.1), Lion(0.04), Rabbit(0.06)).
- But, if the classifier is non-probabilistic, it will only output “Dog”.

Key Aspects of Probabilistic Models

- **1. Random Variables:** Probabilistic models use random variables to represent uncertain events or quantities.
- For example, if we want to predict the probability of rain tomorrow, we can treat "rain" as a random variable with two possible outcomes: "yes" (rain) or "no" (no rain).
- **2. Probability Distributions:** Each random variable in a probabilistic model is associated with a probability distribution, which quantifies the likelihood of different outcomes for that variable.
- Common distributions include:
 - **a. Normal Distribution**
 - Used for continuous values such as **height, weight, marks, temperature**, etc.
 - Many natural phenomena cluster around an average value, with fewer values far from the mean, forming a **bell-shaped curve**.
 - Example: Most students score near average marks, very few score extremely high or low.

- **B. Binomial Distribution**

- Used when an experiment has only **two outcomes**: success/failure, yes/no, pass/fail.
- It gives the probability of getting a certain number of successes in a fixed number of independent trials.
- Example: Tossing a coin 10 times and finding probability of getting 6 heads.

- **C. Multinomial Distribution**

- Extension of binomial distribution when there are **more than two possible outcomes**.
- Used in classification problems with multiple classes.
- Example: Rolling a dice 20 times and counting how many times each face appears.

- **3. Joint and Conditional Probabilities:**

- Joint probability represents the likelihood of multiple events occurring together (e.g., rain and cold weather on the same day).
- Conditional probability captures the probability of one event occurring given that another event has occurred (e.g., the probability of rain given cloudy skies). Conditional probabilities are central to probabilistic reasoning.

- **4. Modeling Dependencies:**

- Probabilistic models are powerful because they can capture dependencies between variables.
- For instance, in Bayesian networks, the dependencies between variables are represented using a graph where nodes represent variables, and edges represent dependencies.

- **5. Learning from Data:** Probabilistic models can learn the parameters of probability distributions from data.

- For instance, in a spam detection model, we can estimate the probabilities of certain words appearing in spam or non-spam emails by analyzing large sets of labeled data. This process is known as parameter estimation.

Types of Probabilistic Models

- **Generative Models:**

These models learn the joint probability distribution of features and labels (e.g., $P(X,Y)$). Generative models can generate new data samples, as they know how both the features and labels are distributed.

- Generative model learns how features are distributed for each class, it can **sample new feature values** from that distribution.
- Example: After learning spam email patterns, it can generate a synthetic spam email containing likely words such as:
Free prize offer now
- Similarly, an image generative model trained on handwritten digits can generate new digit images.
- Common generative models include Naive Bayes, Gaussian Mixture Models, and Hidden Markov Models.
- **Discriminative Models:**
These models focus on learning the conditional probability distribution $P(Y|X)$, where Y is the label, and X is the input data.
- Discriminative models are commonly used for classification, as they focus on directly mapping inputs to outputs. Examples include Logistic Regression, Support Vector Machines, and Neural Networks.

- **Bayesian Models:** Bayesian models incorporate prior beliefs and update these beliefs based on observed data using Bayes' theorem.
- These models are powerful for applications that require quantifying uncertainty.
- For example, if we're trying to predict the average height in a population, we might have a prior belief based on past studies that the average height is around 5'7".
- Examples include Bayesian Linear Regression and Bayesian Networks.
- **Graphical Models:** Graphical models like Bayesian Networks and Markov Random Fields represent variables as nodes in a graph, with edges denoting dependencies. They are used to model complex relationships in structured data.

Naïve Bayes Classifier

- A Naive Bayes classifier is a probabilistic machine learning model that is used for classification task
- It is based on Baye's Theorem

Using Bayes theorem, we can find the probability of A happening, given that B has occurred. The assumption made here is that the predictors/features are independent.

Baye's Theorem

- Bayes theorem works on the principle of Conditional Probability.
- The general statement of Bayes' theorem is **“The conditional probability of an event A, given the occurrence of another event B, is equal to the product of the event of B, given A and the probability of A divided by the probability of event B.**

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- where,
- **P(A)** and **P(B)** are the probabilities of events A and B
- **P(A|B)** is the probability of event A when event B happens
- **P(B|A)** is the probability of event B when A happens

Bayes Theorem Derivation

The proof of Bayes' Theorem is given as, according to the conditional probability formula,

$$P(E_i|A) = P(E_i \cap A) / P(A) \dots (i)$$

Then, by using the multiplication rule of probability, we get

$$P(E_i \cap A) = P(E_i)P(A|E_i) \dots (ii)$$

Now, by the total probability theorem,

$$P(A) = \sum P(E_k)P(A|E_k) \dots (iii)$$

Substituting the value of $P(E_i \cap A)$ and $P(A)$ from eq (ii) and eq(iii) in eq(i) we get,

$$P(E_i|A) = P(E_i)P(A|E_i) / \sum P(E_k)P(A|E_k)$$

- Bayes' theorem is also known as the formula for the **probability of “causes”**. As we know, the E_i 's are a **partition of the sample space S** , and **at any given time only one of the events E_i occurs**. Thus we conclude that the Bayes' theorem formula gives the probability of a particular E_i , given the event A has occurred.

Terms Related to Bayes Theorem

- **Hypotheses:** Events happening in the sample space $E_1, E_2, \dots, E_n$ is called the hypotheses
- **Priori Probability:** Priori Probability is the initial probability of an event occurring before any new data is taken into account. $P(E_i)$ is the priori probability of hypothesis E_i .
- **Posterior Probability:** Posterior Probability is the updated probability of an event after considering new information. Probability $P(E_i|A)$ is considered as the posterior probability of hypothesis E_i .

- **Conditional Probability**

- The probability of an event A based on the occurrence of another event B is termed conditional Probability.
- It is denoted as $P(A|B)$ and represents the probability of A when event B has already happened

- **Joint Probability**

- When the probability of two more events occurring together and at the same time is measured it is marked as Joint Probability. For two events A and B, it is denoted by joint probability is denoted as, $P(A \cap B)$.

- **Random Variables**

- Real-valued variables whose possible values are determined by random experiments are called random variables. The probability of finding such variables is the experimental probability.

Bayes Theorem Examples

- **Example 1:** A person has undertaken a job. The probabilities of completion of the job on time with and without rain are 0.44 and 0.95 respectively. If the probability that it will rain is 0.45, then determine the probability that the job will be completed on time.

Solution:

Let:

- R = Event that it **rains**
- R' = Event that it **does not rain**
- C = Event that the job is **completed on time**

Given probabilities

From the question:

- $P(C | R) = 0.44$ (completion **with rain**)
- $P(C | R') = 0.95$ (completion **without rain**)
- $P(R) = 0.45$

So,

- $P(R') = 1 - 0.45 = 0.55$

Apply Total Probability Theorem

$$P(C) = P(C | R) \cdot P(R) + P(C | R') \cdot P(R')$$

$$P(C) = (0.44 \times 0.45) + (0.95 \times 0.55) = 0.7205$$

- **Example 2:** There are three urns containing 3 white and 2 black balls; 2 white and 3 black balls; 1 black and 4 white balls respectively. There is an equal probability of each urn being chosen. One ball is equal probability chosen at random. what is the probability that a white ball is drawn?

Let:

- $U1, U2, U3$ = choosing urn 1, urn 2, urn 3
- W = drawing a **white ball**

Since each urn is equally likely:

$$P(U1)=P(U2)=P(U3)=1/3$$

Find probability of drawing a white ball from each urn

Urn 1: (3 white, 2 black → total 5)

$$P(W|U1)=3/5$$

Urn 2: (2 white, 3 black → total 5)

$$P(W|U2)=2/5$$

Urn 3: (4 white, 1 black → total 5)

$$P(W|U3)=4/5$$

Apply Total Probability Formula

$$P(W)=P(W|U1)P(U1)+P(W|U2)P(U2)+P(W|U3)P(U3)$$

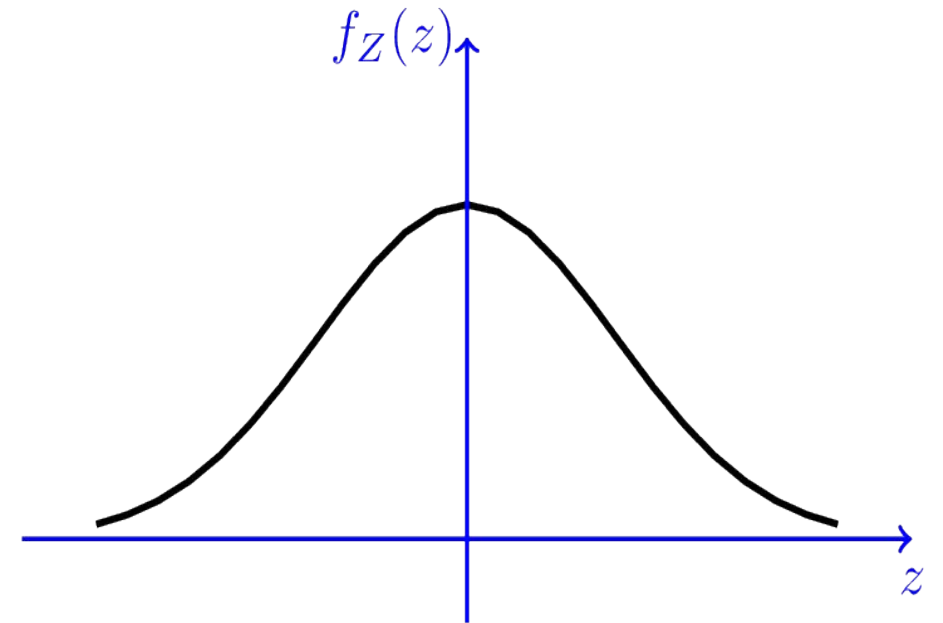
$$P(W)=3/5 \cdot 1/3 + 2/5 \cdot 1/3 + 4/5 \cdot 1/3 = 0.6$$

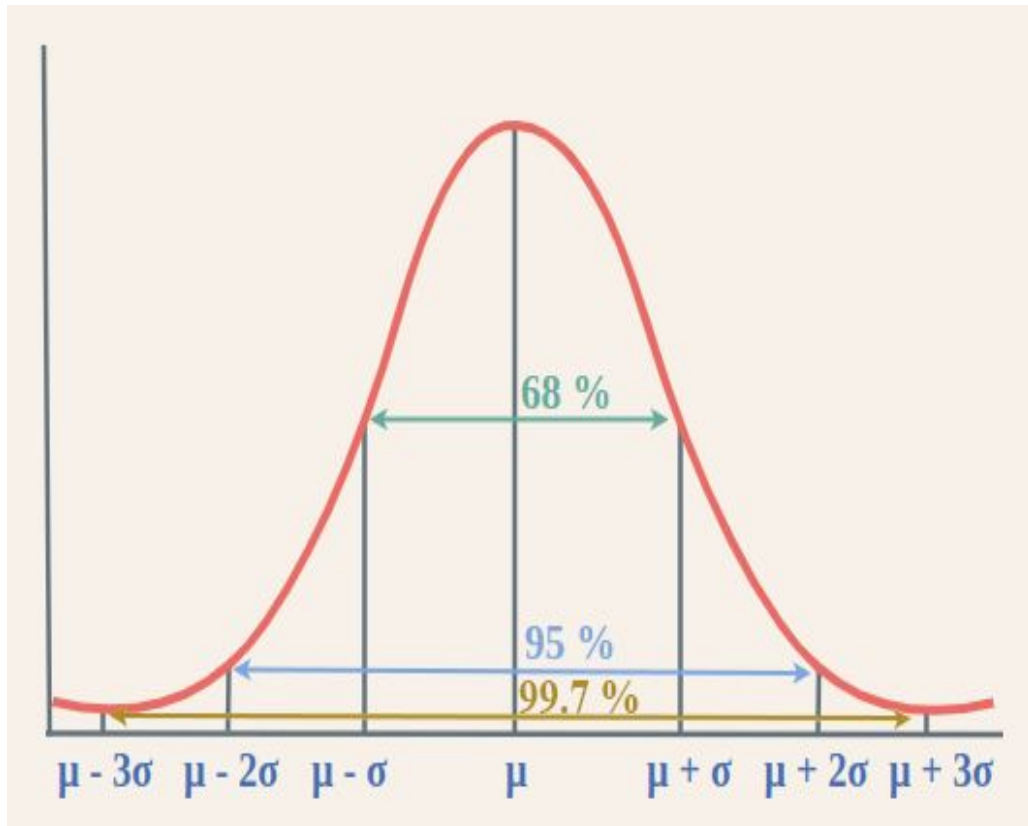
Gaussian Distribution

- The Gaussian distribution, also known as the normal distribution, is a fundamental concept in statistics and probability theory. It describes how the values of a variable are distributed.
- The formula for the Gaussian distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- where:
- x is a value in the distribution.
- μ is the mean.
- σ is the standard deviation.
- $\exp$ denotes the exponential function.





- The standard deviations are used to subdivide the area under the normal curve. Each subdivided section defines the percentage of data, which falls into the specific region of a graph.
- Analysis :** A **smaller standard deviation** results in a **narrower and taller bell curve**, indicating that data points are clustered closely around the mean. Conversely, a **larger standard deviation** leads to a **wider and shorter bell curve**, suggesting that data points are **more spread out from the mean**.
- The **Empirical Rule**, also known as the **68-95-99.7 rule**, quantifies the proportion of data falling within certain intervals around the mean in a normal distribution. It provides a quick way to estimate the spread of data without performing detailed calculations.

Multinomial Distribution

- The multinomial distribution is a generalization of the binomial distribution. While the binomial distribution deals with scenarios where there are only two possible outcomes (success or failure), the multinomial distribution handles situations where there are more than two outcomes.
- The multinomial distribution applies to experiments in which the following conditions are true:
- **Repeated:** The experiment consists of repeated trials, such as rolling a die five times instead of just once.
- **Independent:** Each trial must be independent of the others. For example, if you roll two dice, the outcome of one die does not impact the outcome of the other die.
- **Same probability:** The probability of each outcome must be the same across each instance of the experiment. For example, if a fair, six-sided die is used, then there must be a one-in-six chance of each number being given on each roll.
- **Specific outcome:** Each trial must produce a specific outcome, such as a number between two and 12 if rolling two six-sided dice.

Multinomial Naive Bayes Classifier

- This algorithm is particularly effective for text classification tasks such as spam detection or sentiment analysis.
- Multinomial Naive Bayes classifier assumes that the features (e.g., words in text data) follow a multinomial distribution given the class label. In this case, the probability of observing a particular word in a document is modeled using a multinomial distribution
- *During training, the model estimates **the probabilities of each feature (word) occurring in each class**, and these probabilities are used to predict the class of new documents.*
- For a given class c , the probability of a document with features $x_1, x_2, x_3 \dots x_n$ is given by:

$$P(c \mid x_1, x_2, \dots, x_n) \propto P(c) \prod_{i=1}^n P(x_i \mid c)$$

where $P(x_i \mid c)$ follows a multinomial distribution.

Doc ID	Text	Class
D1	team win match	Sports
D2	player team score	Sports
D3	election vote party	Politics
D4	government vote law	Politics
D5	computer software code	Technology
D6	software internet computer	Technology

We classify a news article into:

- Sports**
- Politics**
- Technology**

We want to classify:

Test Document:

team software win

- Step 1: Build Vocabulary
- Unique word
- Total vocabulary:
- $V=14$

Vocabulary
team
win
match
player
score
election
vote
party
government
law
computer
software
code
internet

- Step 2: Prior Probability

Total documents = 6

Each class has 2 docs.

$$P(Sports) = P(Politics) = P(Technology) = \frac{2}{6} = \frac{1}{3}$$

- Step 3: Word Counts in Each Class

- **Sports Documents**

- D1 = team win match

- D2 = player team score

Word	Count
team	2
win	1
match	1
player	1
score	1

- Total words:

- 6

- **Politics Documents**

- D3 = election vote party
D4 = government vote law

Word	Count
election	1
vote	2
party	1
government	1
law	1

- Total words:
- 6

- **Technology Documents**

- D5:computer software code
- D6:software internet computer

Word	Count
Computer	2
software	2
code	1
Internet	1

- Step 4: Laplace Smoothing

Formula:

$$P(word|class) = \frac{count(word, class) + 1}{6 + 14}$$

Denominator:

$$P(x_i|c) = \frac{count(x_i, c) + 1}{N + k}$$

Where:

- N = total feature count in class
- k = number of possible feature values/classes

20

- **1. Avoids Zero Probability:** No probability becomes zero.
- **2. Handles Unseen Words / Features:** Useful when test data contains words not seen in training for that class.
- **3. Improves Generalization:** Model becomes less overconfident and works better on new data.
- **Step 5: Needed Probabilities for Test Document**
 - Test words:
 - **team software win**

A. Sports

$$P(team|Sports) = \frac{2+1}{20} = 0.15$$

$$P(software|Sports) = \frac{0+1}{20} = 0.05$$

$$P(win|Sports) = \frac{1+1}{20} = 0.10$$

B. Politics

All three absent except smoothing:

$$P(team|Politics) = 0.05$$

$$P(software|Politics) = 0.05$$

$$P(win|Politics) = 0.05$$

C. Technology

$$P(team|Tech) = 0.05$$

$$P(software|Tech) = \frac{2+1}{20} = 0.15$$

$$P(win|Tech) = 0.05$$

- Step 6: Compute Score

Sports

$$\begin{aligned} \text{Score}(\text{Sports}) &= \frac{1}{3}(0.15)(0.05)(0.10) \\ &= 0.00025 \end{aligned}$$

Politics

$$\begin{aligned} \text{Score}(\text{Politics}) &= \frac{1}{3}(0.05)(0.05)(0.05) \\ &= 0.0000417 \end{aligned}$$

Technology

$$\begin{aligned} \text{Score}(\text{Technology}) &= \frac{1}{3}(0.05)(0.15)(0.05) \\ &= 0.000125 \end{aligned}$$

$$\hat{C} = \arg \max_C P(C) \prod_i P(w_i|C)^{x_i}$$

- Highest score = **Sports**

Class	Score
Sports	0.00025
Politics	0.0000417
Technology	0.000125

- **Prediction: Sports**

Binomial Distribution

- A **Binomial Distribution** is a **discrete probability distribution** that gives the probability of obtaining exactly **x successes** in **n independent trials**, where each trial has only **two possible outcomes**:
 - Success / Failure
 - Yes / No
 - Correct / Incorrect
 - Head / Tail
- and the probability of success remains constant in every trial.
- **Conditions for Binomial Distribution**
- A random experiment follows binomial distribution if:
 - **1. Fixed Number of Trials**
 - Number of experiments is fixed = n
 - Example: Toss coin 6 times.
 - **2. Only Two Outcomes Per Trial**
 - Each trial gives:
 - Success
 - Failure
 - Example: Head or Tail

- **3. Trials are Independent:**

- Result of one trial does not affect another trial.
- Example: Tossing a fair coin repeatedly.

- **4. Constant Probability of Success**

- Probability of success remains same in each trial.
- If probability of head is 0.5 in first toss, it remains 0.5 for all tosses.

Let:

X = Number of successes in n trials

Then:

$$X \sim \text{Binomial}(n, p)$$

where:

- n = number of trials
- p = probability of success
- $q = 1 - p$ = probability of failure

Probability Mass Function
(PMF)

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where:

- $x = 0, 1, 2, \dots, n$

$$\binom{n}{x}$$

Number of ways to choose x successes from n trials.

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$p^x$$

Probability of x successes.

$$(1 - p)^{n-x}$$

Probability of remaining failures.

• Problem

We have a dataset of emails with the words they contain and whether each email is labeled as "Spam" or "Not Spam." Here's a sample dataset:

Email	Free	Win	Click	Label
Email1	Yes	Yes	No	Spam
Email2	Yes	Yes	Yes	Not Spam
Email3	Yes	No	No	Spam
Email4	Yes	No	No	Not Spam
Email5	Yes	Yes	Yes	Spam

Now, suppose we receive a new email, **Email6**, that contains the following keywords:

- "free": Yes
- "win": Yes
- "click": No

We want to classify this email as **Spam** or **Not Spam**.

Step 1: Priors

Total emails = 5

- Spam = 3
- Not Spam = 2

$$P(\textit{Spam}) = \frac{3}{5} = 0.6$$

$$P(\textit{NotSpam}) = \frac{2}{5} = 0.4$$

Step 2: Bernoulli Conditional Probabilities

We compute probability of each keyword being Yes/No given class.

For Spam Emails

Spam rows:

1. Yes Yes No
2. Yes No No
3. Yes Yes Yes

Total Spam = 3

Free = Yes

All 3 are Yes:

$$P(\text{Free} = \text{Yes} | \text{Spam}) = \frac{3}{3} = 1$$

Win = Yes

2 of 3:

$$P(\text{Win} = \text{Yes} | \text{Spam}) = \frac{2}{3}$$

Click = No

2 of 3:

$$P(\text{Click} = \text{No} | \text{Spam}) = \frac{2}{3}$$

For Not Spam Emails

Rows:

1. Yes Yes Yes

2. Yes No No

Total = 2

Free = Yes

$$P(\text{Free} = \text{Yes} | \text{NS}) = \frac{2}{2} = 1$$

Win = Yes

$$P(\text{Win} = \text{Yes} | \text{NS}) = \frac{1}{2}$$

Click = No

$$P(\text{Click} = \text{No} | \text{NS}) = \frac{1}{2}$$

Step 3: Posterior for Spam

$$P(\text{Spam}|X) \propto P(\text{Spam}) \cdot P(\text{Free} = \text{Yes}|\text{Spam}) \cdot P(\text{Win} = \text{Yes}|\text{Spam}) \cdot P(\text{Click} = \text{No}|\text{Spam})$$

$$0.6 \times 1 \times \frac{2}{3} \times \frac{2}{3}$$

$$= 0.6 \times \frac{4}{9} = 0.2667$$

Step 4: Posterior for Not Spam

$$P(\text{NS}|X) \propto 0.4 \times 1 \times \frac{1}{2} \times \frac{1}{2}$$

$$0.4 \times 1 \times \frac{1}{2} \times \frac{1}{2}$$

$$= 0.1$$

Step 5: Compare

- Spam score = 0.2667
- Not Spam score = 0.1

Since:

$$0.2667 > 0.1$$

Final Classification

Email6 = Spam

Bernoulli Naive Bayes with Laplace Smoothing

Patient	Fever	Cough	Fatigue	Disease
P1	1	1	1	Yes
P2	1	1	0	Yes
P3	0	0	1	No
P4	1	0	1	Yes
P5	0	1	0	No
P6	0	1	1	No

Where:

- 1 = Present
- 0 = Absent
- Target = Disease (Yes / No)

i) Prior Probabilities

Total patients = 6

- Disease = Yes → 3 patients (P1, P2, P4)
- Disease = No → 3 patients (P3, P5, P6)

$$P(Yes) = \frac{3}{6} = 0.5$$

$$P(No) = \frac{3}{6} = 0.5$$

ii) Conditional Probabilities using Laplace Smoothing

Bernoulli Formula

For binary feature:

$$P(X = 1|C) = \frac{N_1+1}{N_c+2}$$

Since each class has:

$$N_c = 3$$

So denominator:

$$3 + 2 = 5$$

A) For Class = Yes

Patients: P1, P2, P4

Patient	Fever	Cough	Fatigue
P1	1	1	1
P2	1	1	0
P4	1	0	1

Fever

Count(1)=3

$$P(\text{Fever} = 1 | \text{Yes}) = \frac{3+1}{5} = 0.8$$

$$P(\text{Fever} = 0 | \text{Yes}) = 0.2$$

Cough

Count(1)=2

$$P(\text{Cough} = 1 | \text{Yes}) = \frac{2+1}{5} = 0.6$$

$$P(\text{Cough} = 0 | \text{Yes}) = 0.4$$

Fatigue

Count(1)=2

$$P(\text{Fatigue} = 1 | \text{Yes}) = 0.6$$

$$P(\text{Fatigue} = 0 | \text{Yes}) = 0.4$$

B) For Class = No

Patients: P3, P5, P6

Patient	Fever	Cough	Fatigue
P3	0	0	1
P5	0	1	0
P6	0	1	1

Fever

Count(1)=0

$$P(\text{Fever} = 1 | \text{No}) = \frac{0+1}{5} = 0.2$$

$$P(\text{Fever} = 0 | \text{No}) = 0.8$$

Cough

Count(1)=2

$$P(\text{Cough} = 1 | \text{No}) = 0.6$$

$$P(\text{Cough} = 0 | \text{No}) = 0.4$$

Fatigue

Count(1)=2

$$P(\text{Fatigue} = 1 | \text{No}) = 0.6$$

$$P(\text{Fatigue} = 0 | \text{No}) = 0.4$$

iii) Posterior Probability for New Patient

Given:

$$X = (Fever = 1, Cough = 1, Fatigue = 0)$$

For Disease = Yes

$$P(Yes|X) \propto P(Yes) \cdot P(F = 1|Yes) \cdot P(C = 1|Yes) \cdot P(Fa = 0|Yes)$$

$$= 0.5 \times 0.8 \times 0.6 \times 0.4$$

$$= 0.096$$

For Disease = No

$$P(No|X) \propto P(No) \cdot P(F = 1|No) \cdot P(C = 1|No) \cdot P(Fa = 0|No)$$

$$= 0.5 \times 0.2 \times 0.6 \times 0.4$$

$$= 0.024$$

- **iv) Prediction**
- Compare posteriors:
- $P(\text{Yes} | X) = 0.096$
- $P(\text{No} | X) = 0.024$
- **Since $0.096 > 0.024$**
- **Predicted Class:**
- Disease=Yes

Bayesian Belief Network

- A Bayesian network is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph.
- **Graphical Model:** A Bayesian Network is a Directed Acyclic Graph (DAG) where nodes represent random variables and directed edges (arcs) represent dependencies between them.
- **Probabilistic Representation:** It models the joint probability distribution of a set of variables using conditional probabilities, making complex distributions easier to handle.
- **Parent–Child Relationship:** Each node depends only on its parent nodes, and this dependency is quantified using a Conditional Probability Distribution (CPD): $P(X_i | \text{Parents}(X_i))$

- **Conditional Independence:**

If there is no direct path between two nodes, they may be **conditionally independent**, reducing computational complexity significantly.

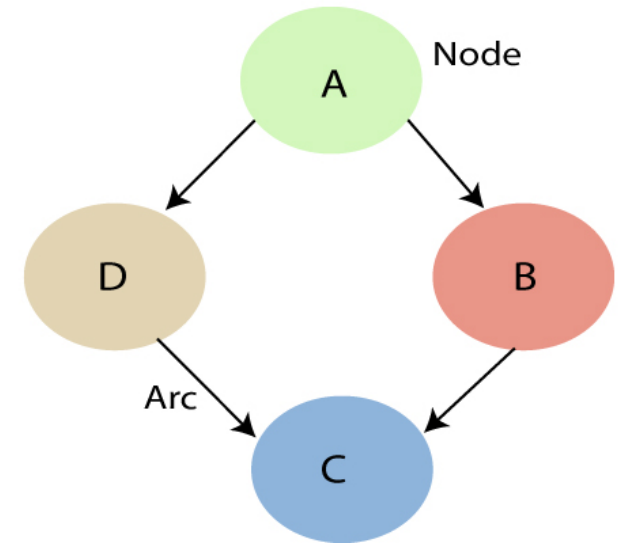
- **Factorization of Joint Probability:**

The overall joint probability can be written as the product of local conditional probabilities:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

- **Applications in Machine Learning:**

Used for **classification, prediction, diagnosis, and decision-making** under uncertainty (e.g., medical diagnosis, spam filtering, fault detection).



Joint probability distribution

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

Key idea behind this equation:

- Instead of depending on all previous variables, each variable depends only on its parents in the graph.
- This is what makes Bayesian Networks efficient and scalable.

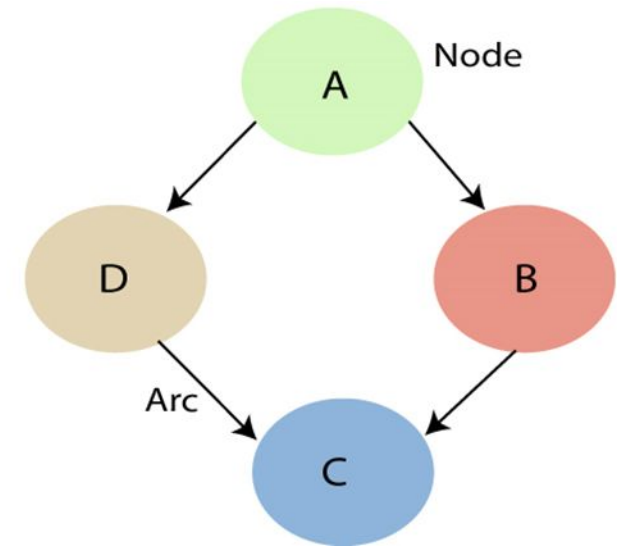
Applying to your given graph (A, B, C, D):

From the structure:

- $A \rightarrow B, A \rightarrow D$
- $B \rightarrow C, D \rightarrow C$

The joint probability becomes:

$$P(A, B, C, D) = P(A) P(B \mid A) P(D \mid A) P(C \mid B, D)$$



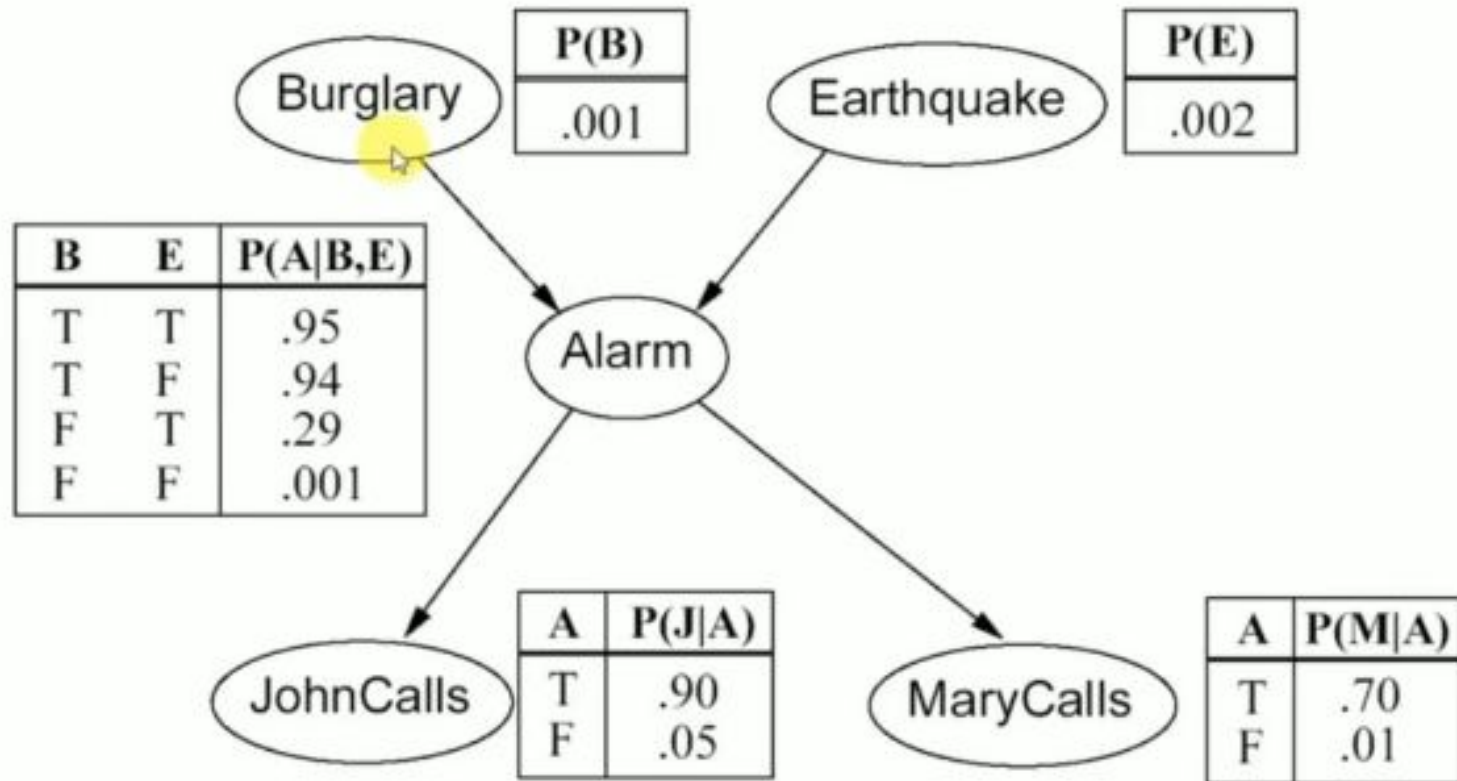
BAYESIAN BELIEF NETWORKS – EXAMPLE – 1

- You have a new burglar alarm installed at home.
- It is fairly reliable at detecting burglary, but also sometimes responds to minor earthquakes.
- You have two neighbors, John and Merry , who promised to call you at work when they hear the alarm.
- John always calls when he hears the alarm, but sometimes confuses telephone ringing with the alarm and calls too.
- Merry likes loud music and sometimes misses the alarm.
- Given the evidence of who has or has not called, we would like to estimate the probability of a burglary.

- **Solution:**

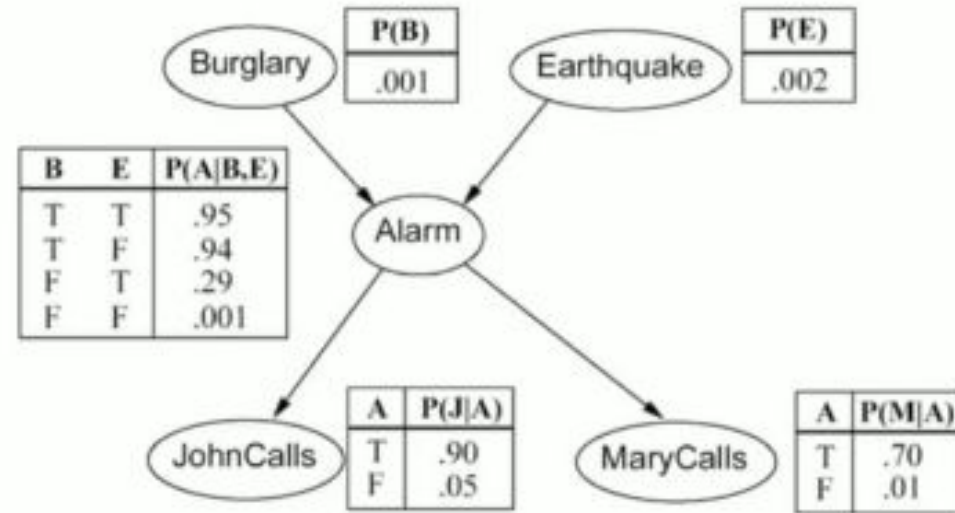
- The Bayesian network for the above problem is given below. The network structure is showing that burglary and earthquake is the parent node of the alarm and directly affecting the probability of alarm's going off, but David and Sophia's calls depend on alarm probability.
- The network is representing that our assumptions do not directly perceive the burglary and also do not notice the minor earthquake, and they also not confer before calling.
- The conditional distributions for each node are given as conditional probabilities table or CPT.
- Each row in the CPT must be sum to 1 because all the entries in the table represent an exhaustive set of cases for the variable.
- In CPT, a boolean variable with k boolean parents contains 2^k probabilities. Hence, if there are two parents, then CPT will contain 4 probability values
- **List of all events occurring in this network:**
 - **Burglary (B)**
 - **Earthquake(E)**
 - **Alarm(A)**
 - **John Calls(J)**
 - **Mary calls(M)**

BAYESIAN BELIEF NETWORKS – EXAMPLE – 1



BAYESIAN BELIEF NETWORKS – EXAMPLE – 1

1. What is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Merry call?



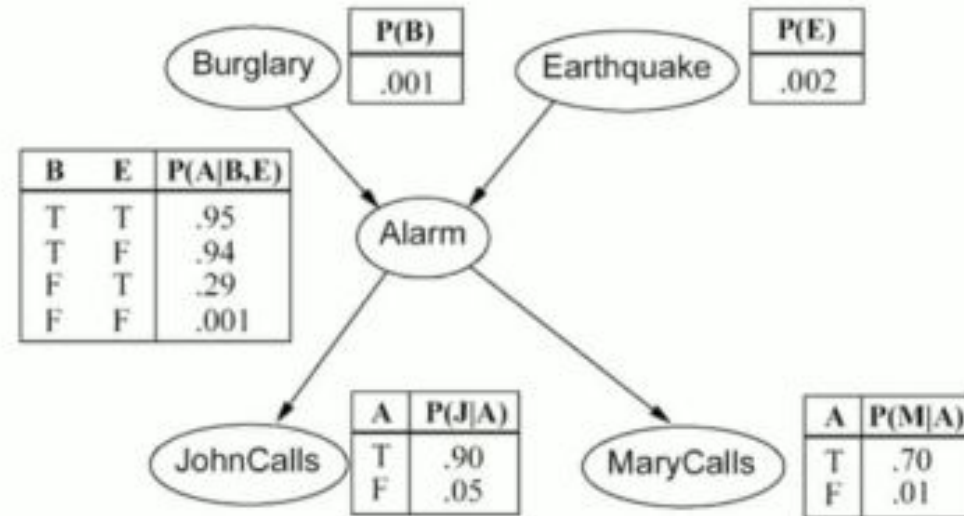
Solution:

$$\begin{aligned} P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) &= P(j \mid a) P(m \mid a) P(a \mid \neg b, \neg e) P(\neg b) P(\neg e) \\ &= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 \\ &= 0.00062 \end{aligned}$$

2. What is the probability that John call?

Solution:

Applying **Law of Total Probability** on the parent node (A)



$$\begin{aligned}
 P(j) &= P(j | a) P(a) + P(j | \neg a) P(\neg a) \\
 &= P(j | a) \{P(a | b, e) * P(b, e) + P(a | \neg b, e) * P(\neg b, e) + P(a | b, \neg e) * P(b, \neg e) + P(a | \neg b, \neg e) * P(\neg b, \neg e)\} \\
 &+ P(j | \neg a) \{P(\neg a | b, e) * P(b, e) + P(\neg a | \neg b, e) * P(\neg b, e) + P(\neg a | b, \neg e) * P(b, \neg e) + P(\neg a | \neg b, \neg e) * \\
 &P(\neg b, \neg e)\} \\
 &= 0.90 * 0.00252 + 0.05 * 0.9974 = 0.0521
 \end{aligned}$$

Markov Model

In a **Markov Model**, the system moves between different states over time, and the key idea is the **Markov property**:

The future state depends only on the current state, not on the past history.

- A **Markov chain** models systems that transition between states, where the next state depends only on the current state (memoryless property).
- The **transition matrix** represents probabilities of moving between states, with each row summing to 1.
- The **Chapman–Kolmogorov equation** is used to compute multi-step transition probabilities by combining individual transitions.

1. Transition Probabilities

These describe how likely it is to move from one state to another.

If we have states $S = \{S_1, S_2, \dots, S_n\}$, then:

- $P(S_j | S_i)$ = probability of moving from state S_i to S_j

These probabilities are usually represented in a **transition matrix**.

Example

Consider a simple weather system with two states:

- Sunny (S)
- Rainy (R)

Transition probabilities:

Current \ Next	Sunny (S)	Rainy (R)
Sunny (S)	0.8	0.2
Rainy (R)	0.4	0.6

2. State Sequences

A state sequence is the actual sequence of states over time.

Example:

- Day 1: Sunny
- Day 2: Sunny
- Day 3: Rainy
- Day 4: Rainy

So the sequence is:

$$S \rightarrow S \rightarrow R \rightarrow R$$

3. Probability of a State Sequence

We can compute the probability of a sequence using transition probabilities.

Example

Let initial probability:

- $P(S) = 0.6, P(R) = 0.4$

Find probability of sequence:

$$S \rightarrow S \rightarrow R$$

Step-by-step:

1. Start in Sunny:

$$P(S) = 0.6$$

2. Transition $S \rightarrow S$:

$$P(S|S) = 0.8$$

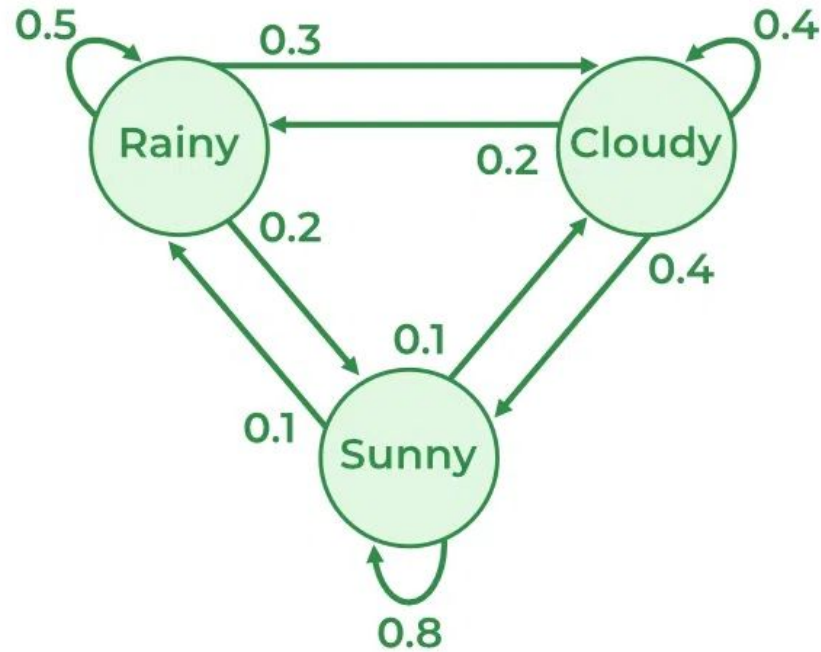
3. Transition $S \rightarrow R$:

$$P(R|S) = 0.2$$

Multiply:

$$P = 0.6 \times 0.8 \times 0.2 = 0.096$$

Problem



Using the transition diagram:

- From Rainy (R) $\rightarrow$ (R=0.5, C=0.3, S=0.2)
- From Cloudy (C) $\rightarrow$ (R=0.2, C=0.4, S=0.4)
- From Sunny (S) $\rightarrow$ (R=0.1, C=0.1, S=0.8)

Assume the initial state is:

$$\pi^{(0)} = [1 \ 0 \ 0] \quad (\text{Day 1 is Rainy})$$

Questions

1. Find the probability of the sequence: R $\rightarrow$ C $\rightarrow$ S
2. Find the probability distribution after **2 days**.

 1. Probability of Sequence $R \rightarrow C \rightarrow S$

Using:

$$P = P(R) \times P(C|R) \times P(S|C)$$

Substitute values:

$$P = 1 \times 0.3 \times 0.4$$

$$P = 0.12$$

 Final Answer:

$$\boxed{0.12}$$

2. State Distribution After 2 Days

Step 1: Transition Matrix

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

Step 2: After 1 Day

$$\pi^{(1)} = \pi^{(0)} \cdot P$$

Since initial is Rainy:

$$\pi^{(1)} = [0.5 \quad 0.3 \quad 0.2]$$

Step 3: After 2 Days

$$\pi^{(2)} = \pi^{(1)} \cdot P$$

Now compute each state:

◆ Probability of Rainy:

$$\begin{aligned} &= 0.5(0.5) + 0.3(0.2) + 0.2(0.1) \\ &= 0.25 + 0.06 + 0.02 = 0.33 \end{aligned}$$

◆ Probability of Cloudy:

$$\begin{aligned} &= 0.5(0.3) + 0.3(0.4) + 0.2(0.1) \\ &= 0.15 + 0.12 + 0.02 = 0.29 \end{aligned}$$

◆ Probability of Sunny:

$$\begin{aligned} &= 0.5(0.2) + 0.3(0.4) + 0.2(0.8) \\ &= 0.10 + 0.12 + 0.16 = 0.38 \end{aligned}$$

✓ Final Answer:

$$\pi^{(2)} = [0.33 \ 0.29 \ 0.38]$$
